Group 7 Project - Milestone 2 (Due Friday, June 27)

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1. Motivation for the idea/description of the problem the application solves

Nowaday people often have to relocate around the US either for new job opportunities or for finding a better living environment. However, single people may prefer locations with higher pay while people with family might want to find areas that are safer for the family. Hence, choosing the ideal location can be difficult as there are many factors to evaluate. We plan to create a useful website for people to look up income, safety and health data based on zip code level to facilitate users on choosing the ideal new home.

2. List of features you will definitely implement in the application

Based on the project proposal and the core functionality, the following features will be definitely implemented:

- **Location Search/ Search Bar:** Users can search for locations (zip codes, cities, states) using a prominent search bar. User will also be able to filter by important metrics such as median_listing_price, count_of_hospital and more
- **Map**: implemented either through Maps or Mapbox API with markers with latitude and longitude equals to the zip code entity attributes (central location of zip code). Clicking a marker opens a modal with a short summary of zip code.
- **Data Visualization of Key Metrics:** Presenting median_listing_price trends over time, counts of public services (hospitals, police departments, fire departments, childcare centers), and health measures.
- **Location Detail Profiles:** Dedicated pages for each zip code/city displaying detailed data.
- **Comparison Tool:** Allow users to select multiple locations (e.g., 2-4) and compare their key metrics side-by-side.
- **Underserved Area Identification:**
    - Identifying ZIP codes with low public safety/health staffing (e.g., police/firefighters per capita) in relation to population.

    - Identifying ZIP codes with a high ratio of population to the number of hospitals, indicating potentially underserved healthcare resources.
- **Home Price Growth Analysis:** Determining the median_listing_price growth for a zip code over the past 5 years.

3. List of features you might implement in the application, given enough time

These features would enhance the application but are considered for a later phase:

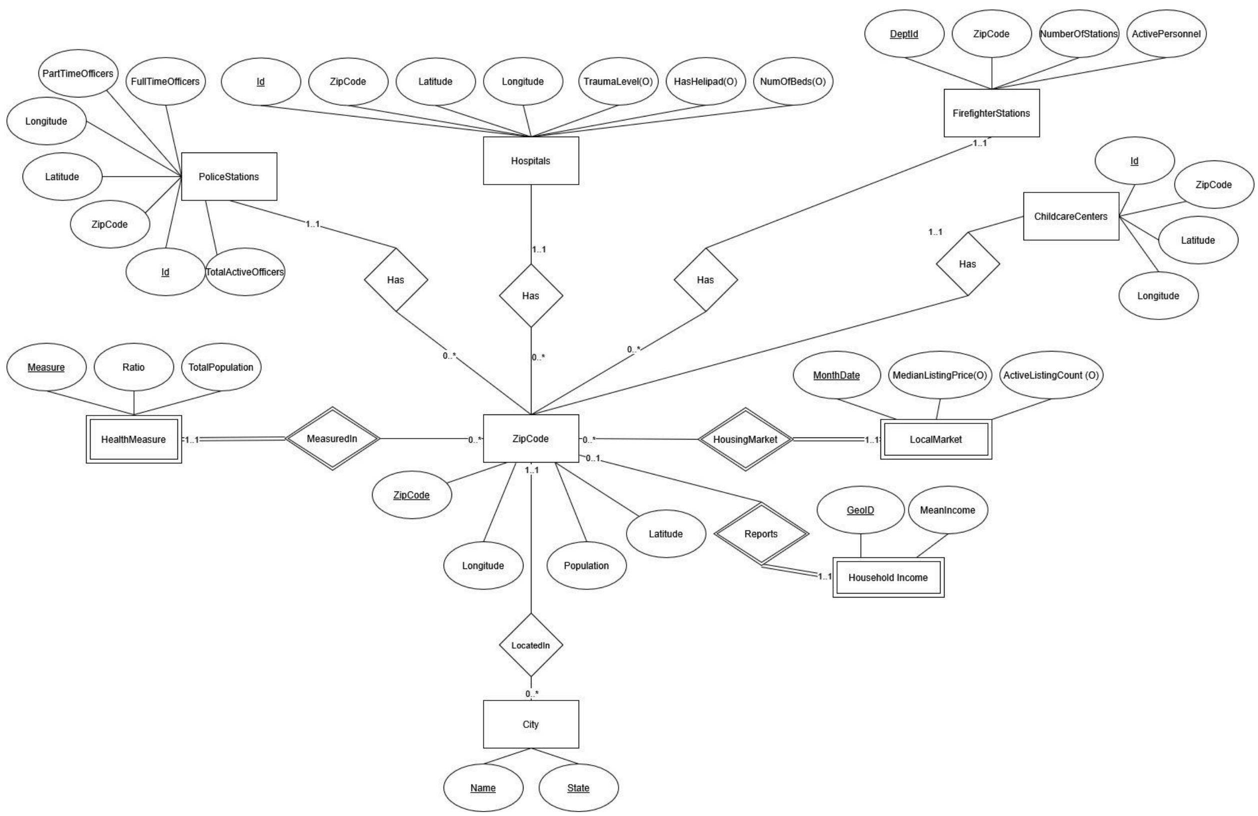
- **Affordability Score/Filter:** Calculate and filter by a ratio of home price to income (requires joining with an income table).
- **Ranking Cities:** Implement a ranking system based on complex ratios like "safety to sale ratio."
- **Growth Hotspots:** Identifying ZIP codes with growth measured by home sales increase, health outcome improvements, and facility increases.
- **Distance to Nearest Hospital:** Calculating the average distance to the nearest hospital by price tier.
- **Job Market Outlook:** Integration with external APIs or datasets to provide job prospect information.
- **School Ratings Integration:** Incorporating data on school quality and ratings.
- **User Accounts & Saved Searches:** Allowing users to save favorite locations and search criteria.
- **User Reviews & Ratings:** Enabling community-driven insights and feedback on locations.
- **Dynamic Data Updates:** More frequent or real-time data integration for certain attributes.
- **AI Price Prediction by zipCode:** Train an AI model from the property data using google cloud services such as BigQuery ML (recommended) or AutoML/VertexAI (more expensive).

4. List of pages the application will have and a 1-2 sentence description of each page.

- **Home Page**: This is the primary entry point where users can initiate their search for a perfect living location using a prominent search bar and quick filter categories, guiding them towards specific interests like "Best for Families" or "Affordable Living."
- **Search Results Page:** This page displays the outcomes of a user's search query in either an interactive map view or a scrollable list view, alongside comprehensive filtering options that allow users to refine results based on various criteria like price, health measures, and public safety.
- **Location Detail Page:** This dedicated page provides an in-depth profile of a single selected zip code or city, showcasing detailed data visualizations across multiple categories such as housing market trends, health statistics, public safety metrics, and local services.

- **Comparison Page:** On this page, users can select and view key metrics of multiple locations side-by-side in a comparative table or card format, enabling direct evaluation and highlighting differences in various attributes.
- **About Us / How We Work Page:** This informational page explains the application's mission, introduces the development team, and transparently details the diverse data sources and methodologies used to generate the insights provided.
- **Contact / Feedback Page:** This page offers a dedicated form for users to submit inquiries, provide suggestions, or offer general feedback regarding the application

5. Relational schema as an ER diagram



6. SQL DDL for creating the database

```
CREATE TABLE City (
	Name VARCHAR(255),
	State VARCHAR(255),
	PRIMARY KEY (Name, State)
)

CREATE TABLE ZipCode (
	ZipCode VARCHAR(5) PRIMARY KEY CHECK (LEN(ZipCode) = 5),
```

```
    Latitude FLOAT NOT NULL,
    Longitude FLOAT NOT NULL,
    Population INT,
    City VARCHAR(255) NOT NULL,
    State VARCHAR(255) NOT NULL,
    FOREIGN KEY (City, State) REFERENCES City (Name, State)
)

CREATE TABLE HealthMeasure (
    Measure VARCHAR(255),
    ZipCode VARCHAR(5),
    Ratio FLOAT NOT NULL CHECK (Ratio >= 0 AND Ratio <= 1),
    TotalPopulation INT NOT NULL,
    PRIMARY KEY (Measure, ZipCode),
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE PoliceStations (
    Id INT PRIMARY KEY,
    ZipCode VARCHAR(5),
    Latitude FLOAT NOT NULL,
    Longitude FLOAT NOT NULL,
    PartTimeOfficers INT NOT NULL,
    FullTimeOfficers INT NOT NULL,
    TotalActiveOfficers INT NOT NULL,
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE Hospitals (
    Id INT PRIMARY KEY,
    ZipCode VARCHAR(5),
    Latitude FLOAT NOT NULL,
    Longitude FLOAT NOT NULL,
    NumOfBeds INT,
    TraumaLevel INT,
    HasHelipad BOOLEAN,
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE ChildcareCenters (
    Id INT PRIMARY KEY,
    ZipCode VARCHAR(5),
    Latitude FLOAT NOT NULL,
    Longitude FLOAT NOT NULL,
```

```
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE FirefighterStations (
    DptId INT PRIMARY KEY,
    ZipCode VARCHAR(5),
    NumberOfStations INT,
    ActivePersonnel INT,
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE LocalMarket (
    MonthDate INT,
    ZipCode VARCHAR(5),
    ActiveListingCount INT NOT NULL,
    MedianListingPrice INT NOT NULL,
    PRIMARY KEY (MonthDate, ZipCode),
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)

CREATE TABLE Household Income (
    GeoID VARCHAR(255),
    ZipCode VARCHAR(5),
    MeanIncome INT
    PRIMARY KEY (GeoID),
    FOREIGN KEY (ZipCode) REFERENCES ZipCode (ZipCode)
)
```

7. Explanation of how you will clean and pre-process the data.

ZipCode entity set, City entity set, and one-many LocatedIn relationship set

These entity sets and this relationship set will be extracted from the United States ZipCode.org dataset.

For the ZipCode entity set, no cleaning or pre-processing is required. We can simply insert the ZIP code, latitude, and longitude columns of the dataset directly into the ZipCode table.

For the City entity set, we will have to extract all unique city + state combinations from the dataset using Pandas (as a given city + state combination often appears more than once in the dataset due to having multiple ZIP codes). All unique combinations will then be inserted into the City table. Additionally, we will include the ZIP code for that city + state combination as a foreign key to the ZipCode table.

Population: Each zip code will be assigned a population count from the US census.
Preprocessing: The table offers a zipcode and total population count without any null values for roughly 33 000 residential zip codes. Using pandas we can extract the information.

HealthMeasure entity set and one-many MeasuredIn relationship set

This entity set and relationship set will be extracted from the CDC Places, Local Data for Better Health (ZCTA Data 2024 release) dataset.

Using Pandas, the primary preprocessing we will have to perform is filtering for a subset of health measures. We will choose 6 measures, which tentatively are

- Obesity among adults
- Depression among adults
- Housing insecurity in the past 12 months among adults
- Feeling socially isolated among adults
- Current asthma among adults
- Visits to doctor for routine checkup within the past year among adults

Additionally, a minor preprocessing step will be for the Ratio attribute. In the dataset, it is measured in percentage. For example, 20 means 20%. We will need to use Pandas to convert the percentage to float such that 0.2 means 20%.

PoliceStations
Source: Homeland Infrastructure Foundation-Level Database

Preprocessing: Using Pandas, the primary preprocessing we will have to perform is filtering for a subset of police stations. We will choose 4 attributes from the table, which tentatively are

- Id is specially assigned to each station
- Zipcode which is assigned to every station (no null entry)
- Latitude, longitude of each station (no null entry)
- Number of part time officers will be calculated as 0 if invalid or null value is present
- Number of full time officers will be calculated as 0 if invalid or null value is present
- Total number of Active Officer calculated as sum of full time and part time active officers. If any number is null then it equates to 0

Hospitals
Source: Homeland Infrastructure Foundation-Level Database

Preprocessing: Using Pandas, the primary preprocessing we will have to perform is filtering for a subset of police stations. We will choose 4 attributes from the table, which tentatively are

- Id is specially assigned to each hospital
- Zipcode which is assigned to every station (no null entry)
- Latitude, longitude of each station (no null entry)
- TraumaLevel will be turned to INT from string in the form LEVEL I, or LEVEL II, or LEVEL III. If value is null or unavailable then it becomes null.
- HasHelipad is a boolean and will be null if no value is provided.
- Total number of beds as reported by each hospital. If no value or invalid entry then it becomes null.

ChildCare Centers
Source: Source: Homeland Infrastructure Foundation-Level Database

Preprocessing: Using Pandas, the primary preprocessing we will have to perform is filtering for a subset of childcare centers. We will choose 4 attributes from the table, which tentatively are

- Id is specially assigned to each childcare center.
- Zipcode which is assigned to every childcare (no null entry)
- Latitude, longitude of each childcare (no null entry)

Firefighter Stations
Source: Source: US Fire Administration:

Preprocessing: Using Pandas, the primary preprocessing we will have to perform is filtering for a subset of police stations. We will choose 4 attributes from the table, which tentatively are

- Id is specially assigned to each department. Could assign an uuid instead of the one in the dataset.
- Zipcode which is assigned to every department (no null entry)
- Total number of stations per department
- Total number of active firefighters calculated as sum of active personnel (Active Firefighters - Career, Active Firefighters - Volunteer, Active Firefighters - Paid per Call). If any number is null then it equates to 0

LocalMarket entity

We will extract this dataset from historical data in Realtor.com for local housing market data.

Preprocessing: We will extract the data from csv into pandas. From within this dataset, we will choose 4 attributes: MonthDate, ZipCode, ActiveListingCount, and MedianListingPirce.

- MonthDate, this is numerical value representing the year and month. There is no null
- ZipCode which is assigned to every MonthDate (no null entry)
- ActiveListingCount, which is the number of the housing listing for this zip code during the corresponding month and year
- MedianListingPrice, which is the median price of the house for this zip during the corresponding month and year

There will be rows where the median listing price and ActiveListingCount will be null as there is no housing market available at that particular zip code. This could be due to military or PO Box zipCode. We will update those row to be null in value for both ActiveListingCount and MedianListingPrice to show there is no data on these rows.

<u>HouseholdIncome entity</u>

We will download the dataset from historical data in US census database for local household Income data.

Preprocessing: We will extract the data from csv into pandas. From within this dataset, we will choose 3 attributes: GeoID, ZipCode, MeanIncome.

- GeoID, which is unique ID corresponding to this area/zipCode
- ZipCode which is assigned to every GeoID (no null entry)
- MeanIncome, which is the Mean income of all households for this zip

There will be rows where the MeanIncome will be null as there is no income available at that particular zip code. This could be due to military zipCode or PO Box zipCode. We will update these rows to be null for income. We will also iterate through zipCode to ensure each row contains 5 characters of numerics.

8. List of technologies you will use. You must use some kind of SQL database.

- PostgreSQL hosted on AWS RDS for database
- Node.js + Express for web server
- React + React Router + undecided styling library for web client
  - Styling library will likely be either CSS-in-JS (emotion or styled-components) or Tailwind CSS plus a matching pre-styled design system kit (in order to move fast)

9. Description of what each group member will be responsible for

We plan to assign team members to each page to give each person the chance to work on both frontend and backend. The current plan is for each page to be done by two people:

Home page - Roy & Dan

Search results page - Dan & Filmon
Location detail page - Filmon & John
Comparison page - John & Roy
About Us - Roy